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TYPES OF ACTIVATION FUNCTIONS AND TERMINOLOGY OF NEURAL NETWORKS.

INTRODUCTION

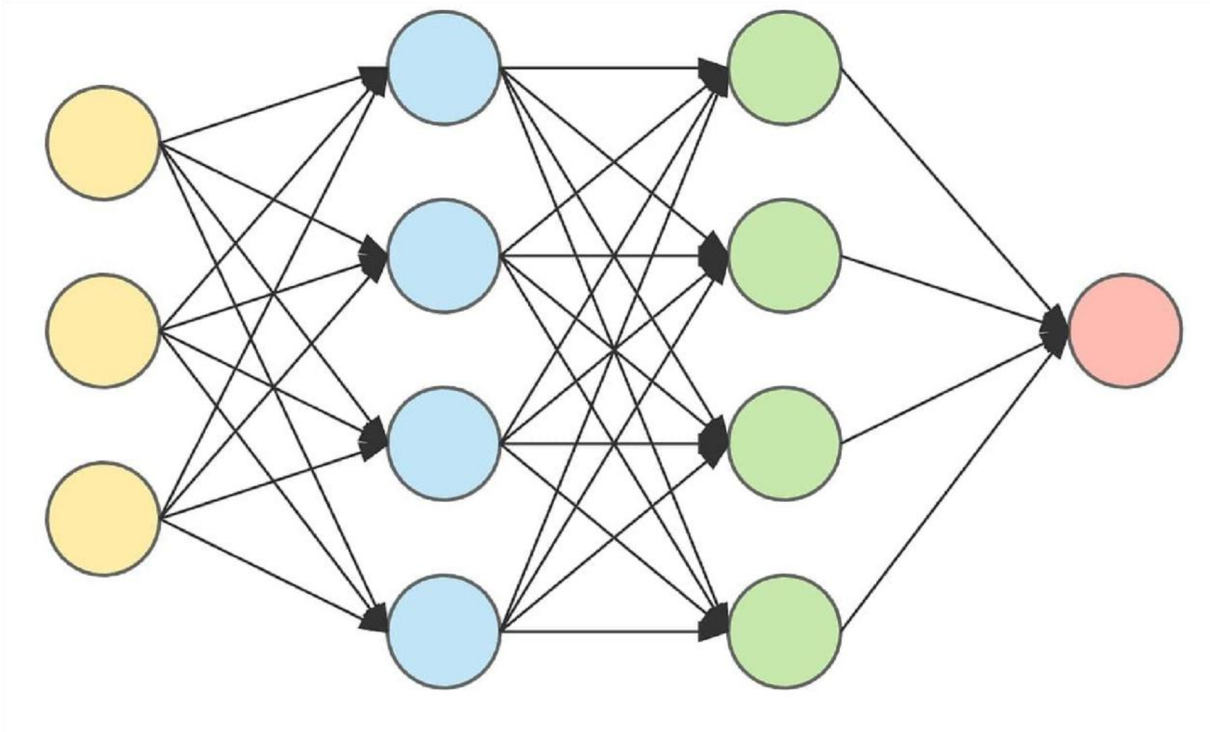
Artificial Intelligence has become an important field in modern technology. Neural Networks are one of the fundamental concepts used in Artificial Intelligence and Machine Learning.

A Neural Network is a computational model inspired by the structure and working of the human brain. It consists of interconnected artificial neurons that process information and learn patterns from data.

Neural Networks are widely used in image recognition, speech recognition, natural language processing, healthcare, and prediction systems.

Activation Functions play an important role in Neural Networks. They determine whether a neuron should be activated and help the network learn complex and non-linear patterns.

ARCHITECTURE OF A NEURAL NETWORK



INPUT LAYER **HIDDEN LAYER 1** **HIDDEN LAYER 2** **OUTPUT LAYER**

A Neural Network is a machine learning model designed to recognize patterns and relationships in data.

It consists of multiple interconnected neurons arranged in different layers.

The main layers of a Neural Network are:

1. Input Layer:

The Input Layer receives data from the outside environment.

Example:

Age, Marks, Salary, Images, etc.

2. Hidden Layer:

The Hidden Layer processes the input data and learns important patterns.

A Neural Network can contain one or more hidden layers.

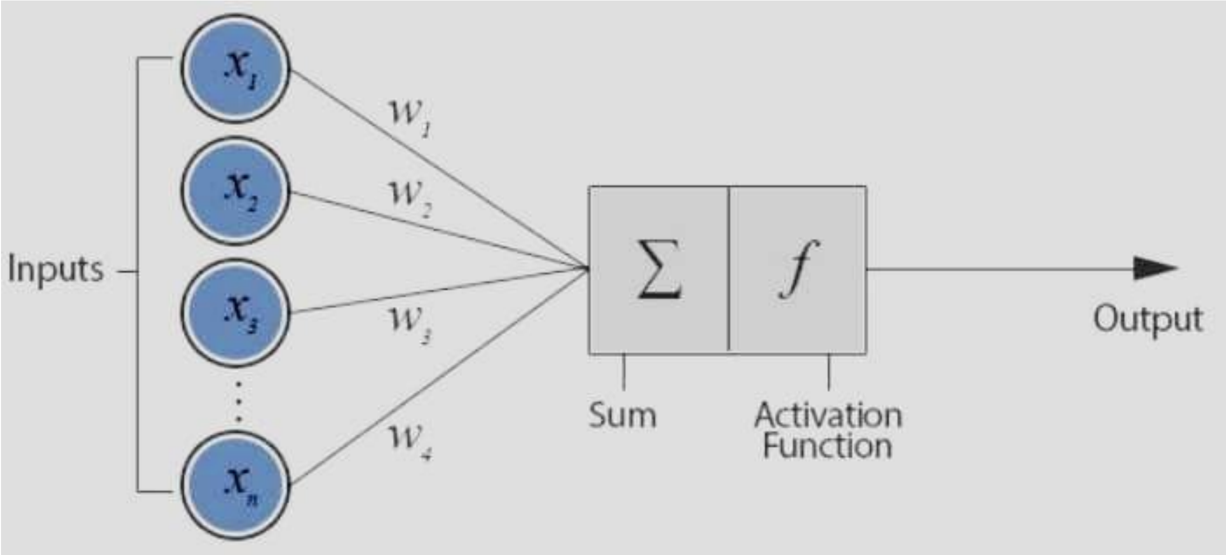
3. Output Layer:

The Output Layer produces the final result or prediction.

Example:

Input → Hidden Layer → Output Layer

TERMINOLOGY OF NEURAL NETWORKS



1. Neuron:

A Neuron is the basic processing unit of a Neural Network. It receives input, processes it, and produces an output.

2. Input:

Input is the data given to the Neural Network for processing.

3. Weight:

Weights determine the importance of each input in the Neural Network.

4. Bias:

Bias is an additional value added to the weighted sum to improve the learning ability of the Neural Network.

5. Activation Function:

An Activation Function decides whether a neuron should be activated or not.

6. Epoch:

One complete pass of the training dataset through the Neural Network is called an Epoch.

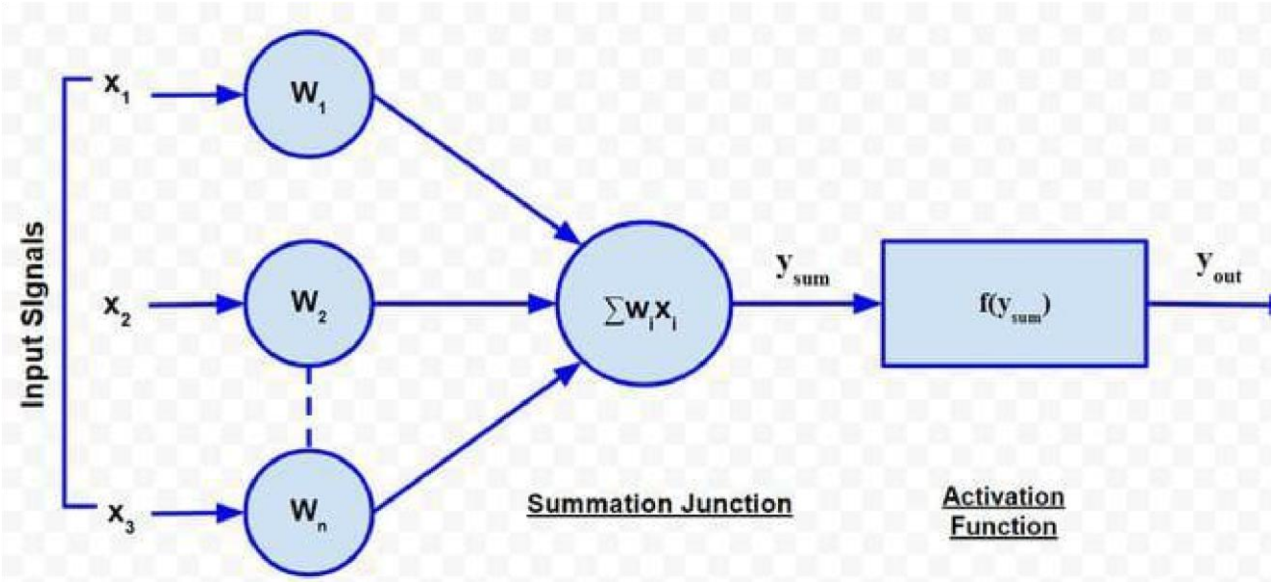
7. Learning Rate:

The Learning Rate determines how much the model updates its weights during training.

8. Loss Function:

A Loss Function measures the difference between the predicted output and the actual output.

ACTIVATION FUNCTION



An Activation Function is a mathematical function used in a Neural Network to determine the output of a neuron.

The main purpose of an Activation Function is to introduce non-linearity into the Neural Network.

Without Activation Functions, a Neural Network would behave like a simple linear model and would not be able to learn complex patterns.

Basic Formula

Output = Activation Function (Σ w x + b)

Where:

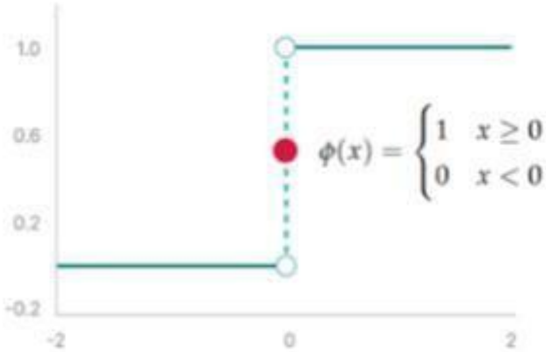
x = Input w

= Weight b

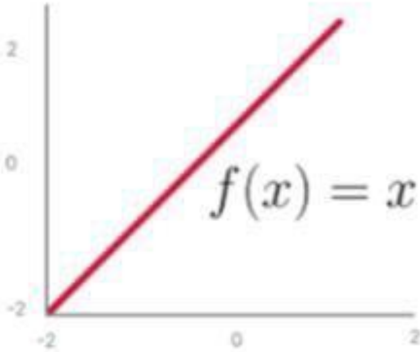
= Bias

Activation Functions help Neural Networks solve complex problems and make accurate predictions.

BINARY STEP FUNCTION



(a) Binary-step function



(b) Linear function

The Binary Step Function is one of the simplest Activation Functions.

It produces only two outputs: 0 or 1.

Working:

If the input value is greater than or equal to zero, the output is 1.

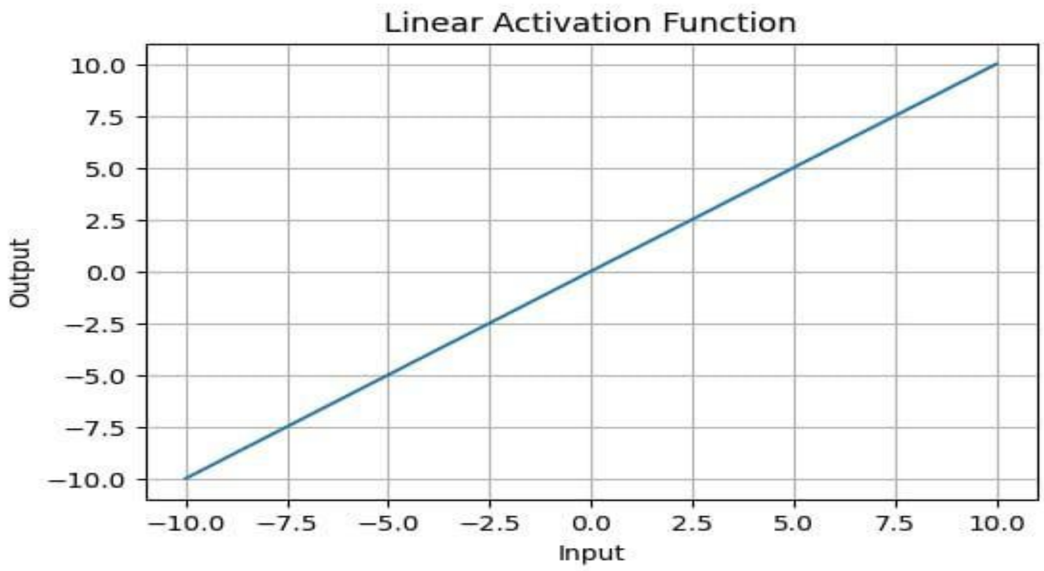
If the input value is less than zero, the output is 0.

Advantages:

* Simple and easy to understand.
* Useful for basic binary decisions.

Disadvantages:

* Cannot handle complex problems.
* Not suitable for deep learning models.



LINEAR ACTIVATION FUNCTION

The Linear Activation Function produces an output that is directly proportional to the input.

Formula

f(x) = x

Advantages

* Simple to implement.

* Useful for regression problems.

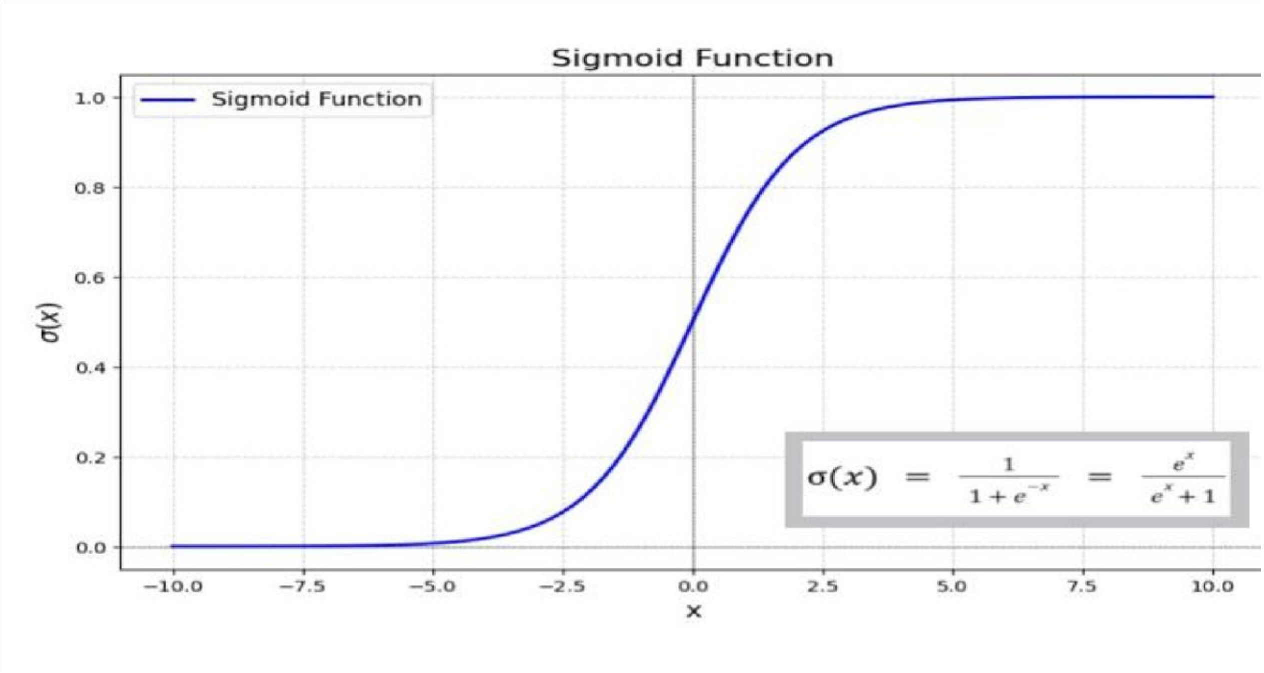
Disadvantages

* Cannot learn complex non-linear patterns.

* Not suitable for hidden layers in deep Neural Networks.

Application

Linear Activation Functions are commonly used in the output layer for regression problems.



SIGMOID FUNCTION

The Sigmoid Activation Function converts input values into values between 0 and 1.

Formula

f(x) = 1 / (1 + e $^{-x}$)

Output Range

0 to 1

Advantages

* Produces smooth output.

* Useful for probability prediction.

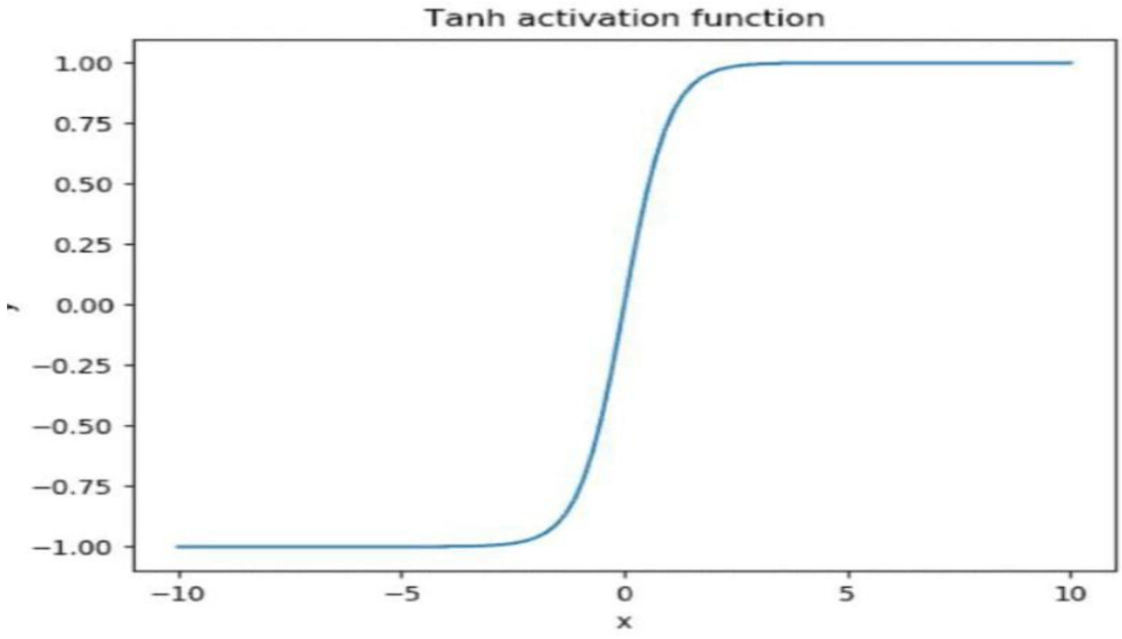
Disadvantages

* Can cause the vanishing gradient problem.

* Training can become slow.

Application

Sigmoid is commonly used for binary classification problems.



TANH FUNCTION

The Tanh Function is similar to the Sigmoid Function but produces values between -1 and +1.

Formula

f(x) = tanh(x)

Output Range

-1 to +1

Advantages

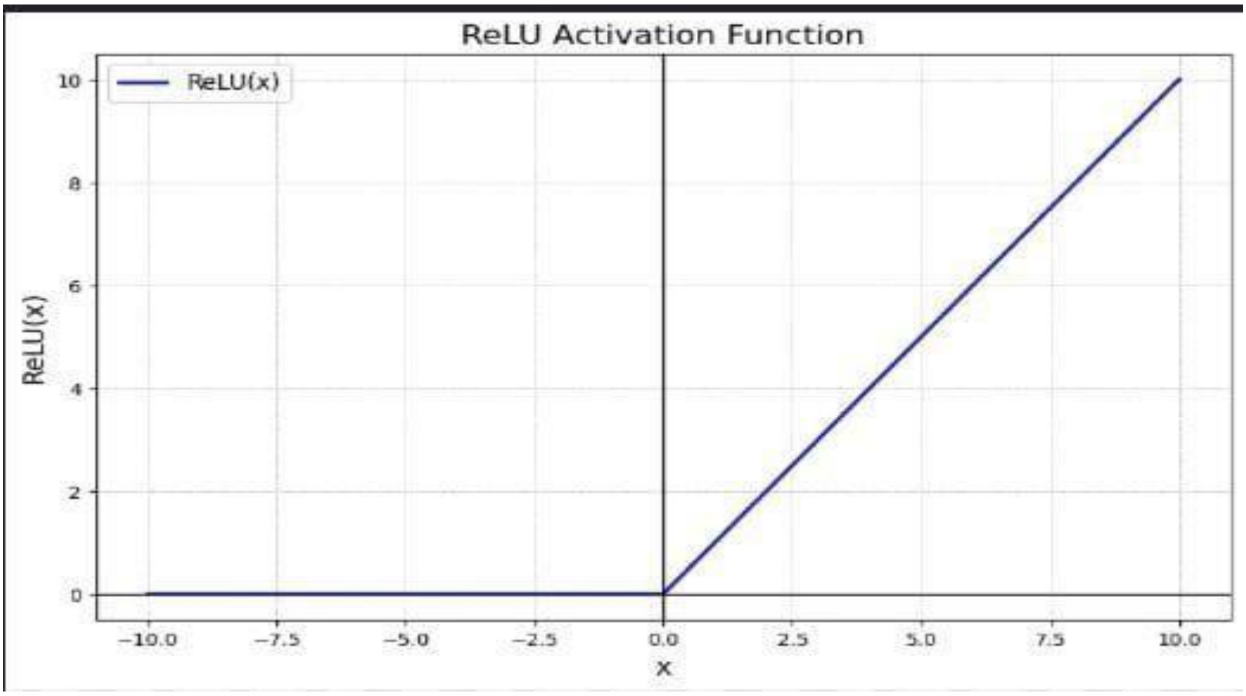
* Can perform better than Sigmoid in some cases.

Disadvantages

* Can also suffer from the vanishing gradient problem.

Application

Tanh is commonly used in hidden layers of Neural Networks.



Re LU FUNCTION

Re LU stands for Rectified Linear Unit.

It is one of the most commonly used Activation Functions in Deep Learning.

Formula

f(x) = max (0, x)

Working

If the input is negative, the output is 0.

If the input is positive, the output is equal to the input.

Advantages

* Simple and fast.
* Helps Neural Networks learn faster.

Disadvantages

APPLICATIONS OF NEURAL NETWORK

Neural Networks are used in many real-world applications.

1. Image Recognition

Neural Networks can identify objects, faces, and patterns in images.

2. Speech Recognition

They are used in voice assistants and speech-to-text applications.

3. Healthcare

Neural Networks help in disease prediction and medical image analysis.

4. Natural Language Processing

They are used for language translation, chatbots, and text analysis.

5. Self-Driving Cars

Neural Networks help vehicles identify roads, objects, and traffic signals.

CONCLUSION

Neural Networks are an important part of Artificial Intelligence and Machine Learning. They are capable of learning complex patterns from data and making predictions.

Activation Functions play a major role in the performance of Neural Networks. Different Activation Functions are used depending on the type of problem and the architecture of the Neural Network.

Functions such as Sigmoid, Tanh, Re LU, Leaky Re LU have different advantages and applications.

Therefore, selecting an appropriate Activation Function is important for building an efficient and accurate Neural Network model.